WORKED SOLUTION

# STEP III 2011, Question 9

Mechanics • Motion on a Cylinder

**STEP III 2011, Question 9**

Read the question before you go on. It is in the STEP Support Programme's question database:

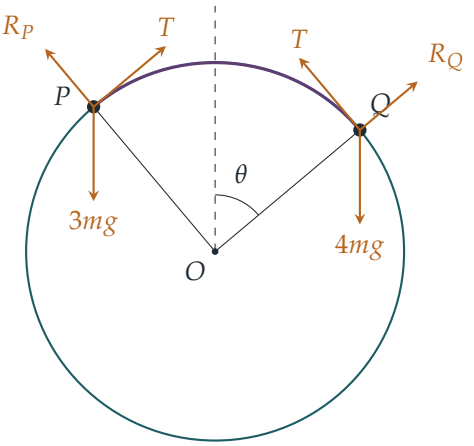
step.maths.org/questions/11-s3-q9

## The key idea

A string of length $\frac{1}{2}\pi a$ lying on the cylinder subtends a right angle at $O$, so $OP$ and $OQ$ are always perpendicular and the whole system has a single coordinate, $\theta$. Energy conservation then gives $\dot{\theta}^2$ directly, and Newton's second law towards the centre gives each normal reaction in terms of $\dot{\theta}^2$. Losing contact means the normal reaction falls to zero. For the tension, use the tangential equations: adding them eliminates $T$ and gives $\ddot{\theta}$, and either one alone then gives $T$.

## Solution

**Model.** $P$ and $Q$ are particles, the string is light and inextensible, and the cylinder is smooth. A light string over a smooth surface has the same tension $T$ throughout. The arc between the particles has length $\frac{1}{2}\pi a$, so $\angle POQ = \frac{1}{2}\pi$. Measure $\theta$ as the angle $OQ$ makes with the upward vertical, with $Q$ on one side of the top. Then $OP$ makes angle $\frac{1}{2}\pi - \theta$ with the upward vertical on the other side. While the particles stay on the surface, both move on the circle of radius $a$ with angular speed $\dot{\theta}$, so each has speed $a\dot{\theta}$.



**Equilibrium.** Resolving along the surface at each particle (the normal reaction has no tangential component, and the string pulls tangentially towards the top), for $Q$ we get $T = 4mg\sin\theta$ and for $P$ we get $T = 3mg\sin(\frac{1}{2}\pi - \theta) = 3mg\cos\theta$. So initially $\tan\theta = \frac{3}{4}$, that is, $\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$.

**Energy.** Take $O$ as the level of zero potential energy. $Q$ is at height $a\cos\theta$ and $P$ at height $a\cos(\frac{1}{2}\pi - \theta) = a\sin\theta$. The normal reactions do no work (they're perpendicular to the motion), and the tension does no net work on the system (the string is inextensible), so energy is conserved. The system starts from rest, as the disturbance is slight, so

$$\tfrac{1}{2}(3m + 4m)a^2\dot\theta^2 + 4mga\cos\theta + 3mga\sin\theta = 4mga\cdot\tfrac{4}{5} + 3mga\cdot\tfrac{3}{5} = 5mga.$$

Multiplying by $2/(ma)$ gives

$$7a\dot\theta^2 + 8g\cos\theta + 6g\sin\theta = 10g,$$

as required. As $Q$ moves down, $\theta$ increases from $\tan^{-1}\frac{3}{4}$.

#### Part (i)

$Q$ moves on a circle of radius $a$ with speed $a\dot\theta$, so it has acceleration $a\dot\theta^2$ towards $O$. The component of its weight towards $O$ is $4mg\cos\theta$, and $R_Q$ acts outwards. The string is tangential, so it contributes nothing radially. Hence

$$4mg\cos\theta - R_Q = 4ma\dot\theta^2, \qquad \text{so} \qquad R_Q = 4mg\cos\theta - 4ma\dot\theta^2.$$

$Q$ leaves the surface when $R_Q$ falls to zero, that is, when $a\dot\theta^2 = g\cos\theta$. Substituting this into the energy equation at $\theta = \beta$:

$$7g\cos\beta + 8g\cos\beta + 6g\sin\beta = 10g \quad\Longrightarrow\quad 15\cos\beta + 6\sin\beta = 10,$$

as required.

**?** **Ask yourself** Why does the energy equation still hold right up to $\theta = \beta$?

It holds as long as both particles are on the cylinder, and the question tells us $Q$ leaves first. You can also check this. The same argument for $P$ gives $R_P = 3mg\sin\theta - 3ma\dot\theta^2$, which vanishes when $13\sin\theta + 8\cos\theta = 10$. At $\beta \approx 73.6^\circ$ the left-hand side is about 14.7, so $R_P$ is still positive.

#### Part (ii)

Now resolve tangentially, taking the direction of motion as positive. $Q$ moves away from the top with tangential acceleration $a\ddot\theta$, and $P$ moves towards the top with the same tangential acceleration $a\ddot\theta$ (the string keeps their speeds equal). For $Q$, gravity has component $4mg\sin\theta$ forwards and the tension pulls back:

$$4ma\ddot\theta = 4mg\sin\theta - T. \tag{1}$$

For $P$, the tension pulls it forwards and gravity has component $3mg\cos\theta$ backwards:

$$3ma\ddot{\theta} = T - 3mg\cos\theta. \tag{2}$$

Adding (1) and (2) eliminates $T$:

$$7ma\ddot{\theta} = 4mg\sin\theta - 3mg\cos\theta.$$

(This agrees with differentiating the energy equation with respect to time and dividing by $2\dot{\theta}$.) Substituting into (1):

$$\begin{aligned} T &= 4mg\sin\theta - 4ma\ddot{\theta} \\ &= 4mg\sin\theta - \tfrac{4}{7}m(4g\sin\theta - 3g\cos\theta) \\ &= \tfrac{12}{7}mg\sin\theta + \tfrac{12}{7}mg\cos\theta = \tfrac{12}{7}mg(\sin\theta + \cos\theta), \end{aligned}$$

as required. This is positive for $0 < \theta < \frac{1}{2}\pi$, so the string stays taut throughout.

## Checks

- At equilibrium, $\sin\theta + \cos\theta = \frac{7}{5}$, so $T = \frac{12}{5}mg$. This matches $4mg \cdot \frac{3}{5}$ and $3mg \cdot \frac{4}{5}$.
- Writing $15\cos\beta + 6\sin\beta = \sqrt{261}\cos(\beta - \alpha)$ with $\tan\alpha = \frac{2}{5}$ gives $\beta \approx 1.2839$ rad $\approx 73.56°$. Stepping the motion numerically from just past equilibrium, we found $R_Q$ first reaches zero at this same angle, while $R_P$ stays positive. The energy relation and the tension formula held along the whole path to within rounding error.

## Common slips

- Taking $P$'s height as $a\cos\theta$ rather than $a\sin\theta$. The right angle $\angle POQ$ is the whole point of the string's length.
- Giving each particle a different tension, or putting $T$ into the radial equation. The string touches the particle tangentially, so $T$ appears only in the tangential equations.

## Beyond STEP

With one coordinate $\theta$ describing everything, this is a one-degree-of-freedom system. Lagrangian mechanics gets the equation of motion straight from the kinetic and potential energies, with no forces of constraint at all. The normal reactions and tension then come out as the "Lagrange multipliers" of the constraints. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to deepen your feel for constrained motion and energy methods before meeting Lagrange.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.